Joint sign guards and Pell descent for smaller consistency equations

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Abstract

This iteration combines sign recovery, calibrated native guards, simultaneous positivity, and carefully ordered elimination of arithmetic coordinates. The resulting unrestricted-integer family includes a quartic in 37 unknowns, a sextic in 31, an octic in 25, and polynomials in 23, 22, 21, 20 and 19 unknowns at degrees 12, 16, 20, 32 and 88 respectively. The important gain is not a new universal-machine bound: it is that joint sign compression and zero-set row reduction allow many more source coordinates to disappear without introducing rational witnesses. We give complete fixed coefficient bindings and compact defining equations.

Independently, adjacent-coefficient Pell descent removes a denominator guard from the eleven-positive-variable construction, lowering degree 325 to 277 and operations 113 to 111. A centered transport gives degree 257 in 114 operations, a further degree/cost tradeoff. The direct proof-checker route also gains smaller exact local polynomial graphs, and the inert-prime certificate no longer needs a CRT index variable or a second variable-power graph. Those compiler-interface advances are not presented as completed small consistency equations.

All endpoint arguments are ordinary mathematics with explicit inherited Pell, native compiler and refutation-calibration dependencies. Finite replays and independent symbolic checks supplement those arguments; they do not constitute a new end-to-end Lean/Rocq certificate or a proof of consistency. No global minimality or literature-wide priority is claimed.

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1 The target and the distinction between an equation and an interface

We seek a fixed polynomial $P \in \mathbb{Z}[x_1, \ldots, x_n]$ and a stated domain D for which

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

Here $\mathbb{N}$ includes zero; $\mathbb{N}_{>0}$ and $\mathbb{Z}$ are different domains. A polynomial register is an abbreviation, not a hidden witness. Fixed coefficient integers are specified independently of consistency, by complete finite recipes and authenticated payloads. None of our defining equations contains an unknown exponent, factorial or binomial coefficient. The alternative CRT interface does use such functions and is explicitly excluded from the completed-polynomial rankings.

The constructions are compared along several axes: domain, unknown count, degree, arithmetic operations, coefficient payload and proof burden. An unrestricted-integer equation is not dominated by a positive-domain equation merely because their nominal variable counts agree. Likewise a reduction in variables at the cost of degree is a tradeoff, not an unqualified simplification of every representation. Our principal completed family uses the coherent source34 tuple and two additional powers of two. The positive Pell family uses compact44 instead. No individual coefficient is exchanged between these tuples.

The fresh starting point is repository commit `2605669a532342f153b13eb6b4a0af4b54f5fa37`. All eight original wave-ten archives are pinned and retained unchanged. The ordinary proofs establish the reductions relative to the named inherited results, not a newly checked derivation in the repository's encoded ZFC calculus. This distinction does not turn a finite test into a universal proof, nor does it assume the truth of Con(ZFC).

2 Fresh baseline and proof boundary

The immutable repository checkpoint is `2605669a5323`; the eight wave-ten originals are pinned to arrival `9513bd88cefb`. We checked all 1,183 release-manifest entries and replayed the eight roots normally and seven in optimized Python, always in fresh temporary copies. All fifteen replays passed, and every original remained unchanged. A separate reader verified topology, liveness and operation counts for 21 literal circuits. These are finite algebra/data checks, not a new proof-assistant build or an internal consistency derivation.

Domain and family	Variables	Degree	Operations
Integer, cone or determinant	38	4	not optimized
Integer, cone or determinant	32	6	not optimized
Integer, cone or determinant	27	8	not optimized
Integer, nested positivity	26	16	not optimized
Integer, free Lucas	10	1,576,864	481
Positive, root transfer	11	325	113
Positive, rational Pell descent	12	82	102
Positive, centered product	12	110	100
Positive, centered SOS	12	164	99
Positive, strong congruence	13	28	98
Positive, adaptive carrier	13	30	92
Positive, retained product	14	24	101
Positive, composed nine	9	1,379,730	478
Positive, repository H2082	9	1,845,940	443

This is a selected domain-preserving comparison, not a global optimality claim. An operation is one live binary addition, subtraction or multiplication; fixed numeral leaves and register reuse are free. The cone and determinant equations have the same sizes but different polynomials. Their six fixed payloads have respectively 881,696 and 615,883 raw bytes. Neither licenses arbitrary substitution of the smaller compact54 tuple.

The latest formal checkpoint completes the supplied numeric sealed-ZFC refutation checker with 83,281 raw and 76,313 optimized program-tree nodes. Those are source-syntax counts, not polynomial sizes. Native arithmetic now extracts sparse assignments and the local clause bands, but whole-payload binding, phase synchronization and global native calibration remain open. The internal semiring, local wire quartics, sum-of-squares conjunction and literal tree-to-quartic compilation are checked on all internal naturals. The full checker transport on nonstandard proof codes and the final object-language consistency/unsolvability biconditional remain open.

3 Commuting restrictions: 37 integers at degree four

The degree-four, degree-six and degree-eight unrestricted-integer endpoints improve from the inspected incoming sizes $38, 32, 27$ to

$$\boxed{37 \text{ variables, degree } 4}, \qquad \boxed{31 \text{ variables, degree } 6}, \qquad \boxed{26 \text{ variables, degree } 8}.$$

This is a compatibility result: source 70's residue orientation and source 71's calibrated determinant guard restrict the *same* canonical source-34 witness family, so their savings combine. Neither ingredient alone is new here. The conclusions use the explicitly inherited unbounded Pell, native compiler and syntactic-calibration theorems. They are not new kernel-checked consistency biconditionals, and no global minimality is claimed.

3.1 The explicit 37-variable quartic

Use the authentic fixed tuple (β, K_0, M_C, M_F), with $\beta = 2^{1064921} - 1$ and M_F already shifted by β. The only new fixed numerals are

$$A_0 = 2^{1113}, \qquad B_0 = 2^{666104}.$$

All six integers are regenerated from their recipes and actual payload bytes; their total minimal unsigned size is 615,883 bytes. Let

$$q = \beta J{+}1, \quad C = q{-}F{-}\alpha, \quad X = 3X_3{-}1, \quad H = 4a{+}3, \quad k = \eta{+}\zeta, \quad \mathrm{Pos}(x, y, z) = 1{+}x^2{+}y^2{+}z(z{+}1).$$

These are polynomial abbreviations, not additional unknowns. The 37 unrestricted integer unknowns are

$$J, F, \alpha, z, f, h, r, v, w, \tau, \eta, \zeta, y, \gamma, o, j, N, E, P, c, a, X_3, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta,$$
$$e_1, e_2, e_3, t_1, t_2, t_3.$$

Here v, w abbreviate the JSON names `v_scale,w_scale`, e_1, e_2, e_3 the retained $\eta\zeta$ positivity coordinates, and t_1, t_2, t_3 the new guard coordinates. Define the following 24 residuals by subtracting each right side from its left side:

$$\begin{aligned}
L &= q^2 - 1, & N &= q(L+1), & X &= 4qw,\\
a - E &= (2v+1)N, & E(X+1) &= Xa, & c &= (a-E)k + \eta,\\
P &= k + E(c-\eta), & \tau^2 &= P^2 - Pk + 1, & D &= X + ac + \gamma H,\\
\Delta &= a^2 + 4a + 3, & C_2 &= c^2, & D^2 - \Delta C_2 &= 1,\\
F_2 &= f^2, & V + c &= fo, & V - cj + k - 1 &= hE,\\
g &= y^2 - V^2, & I &= (2r+1)C_2, & B_{\mathrm{aux}} &= I^2,\\
B_{\mathrm{aux}} &= \Delta(F_2 - 1), & y^2 - B_{\mathrm{aux}} g &= 1.
\end{aligned}$$

The remaining four equations are

$$\begin{aligned}
0 &= k - hE + FN + qL - qN + q^2 - 1 - F(q+L)\\
&\quad - \alpha L - (M_C + qM_F)J,\\
(K_0 + 4w)C &= F + z(q-1),\\
\eta\zeta &= \mathrm{Pos}(e_1, e_2, e_3),\\
(A_0F)(B_0C) - (q - A_0F - B_0C)^2 &= \mathrm{Pos}(t_1, t_2, t_3).
\end{aligned}$$

The polynomial is the sum of these 24 squared residuals. All residuals are quadratic or lower in the displayed unknowns, and the coefficient of τ^4 is exactly one. Thus the polynomial has exact degree four after authentic coefficient specialization. Its complete expanded residual list is `determinant37_quartic_direct.json`.

3.2 Arbitrary-signed soundness, before canonicality

Put $U = A_0F$ and $V_0 = B_0C$. The determinant row gives $UV_0 > (q - U - V_0)^2 \geq 0$. Consequently U, V_0 have the same nonzero sign and

$$|q - U - V_0| < \sqrt{UV_0} \leq \frac{|U| + |V_0|}{2}.$$

Thus q, F, C have one sign, and $|q| > (A_0|F| + B_0|C|)/2 \geq |F| + |C|$ forces α to share it. This restores both old product guards using two fresh Pos triples. No canonical-witness or positive-radix assumption has been made.

It remains to restore the deleted square witness, not to assume it. Set $Q = |q|$, $Y = (2v+1)q^3$ and R equal to the authentic packing index. The retained lift yields

$$E = XY, \quad a = (X+1)Y, \quad c = kY + \eta, \quad k = R + 1 + hE, \quad V = of - c = jc - R.$$

The common-sign packing estimates imply $Q \geq 16$ and $Q(Q^2-1) < R < Q^4 - 2$. Since $X = 3X_3 - 1 \neq 0$ and $X = 4qw$,

$$|X| \geq 4Q, \qquad |Y| \geq Q^3, \qquad |E| \geq 4Q^4 > 3R + 1.$$

Write $B = |a+2|$, $P_0 = |2XY^2+1|$, $c_0 = |c|$, $K = |k|$. The first and main Pell norms, valid for either sign of X, give

$$K = 2\psi_{P_0}(n), \qquad c_0 = \psi_B(p), \qquad p > n.$$

Here $P_0 > B > R$ and $K(|Y|-1) < c_0 < K(|Y|+1)$. The index congruence is

$$\operatorname{sgn}(k)\operatorname{sgn}(X)^{n-1}\,2n \equiv R+1 \pmod{|E|}.$$

Its positive branch has $n \geq (R+1)/2$ and its negative branch has $n > R$. These estimates establish the inherited ordinary strong-rank hypotheses. The signed auxiliary step-down then gives $p = R$ without choosing any sign. The negative branch is impossible, and no wrap is possible in the other: $R = 2n-1$ is odd.

For $U_0 = |X| \geq 64$, $Z_0 = |Y| \geq 4096$ and $r_0 = (R-1)/2$, Pell growth gives

$$\frac{c_0}{K} \geq \frac{1}{2}\left(\frac{(2B-1)^2}{2P_0}\right)^{r_0} > \frac{(U_0-3)^{r_0}}{2}.$$

The strict inequality follows from the universally positive difference

$$(2(U_0-1)Z_0-5)^2 - (U_0-3)(4U_0Z_0^2+2) > 0.$$

After $U_0 = 64+u$, $Z_0 = 6+z$ it is $4uz^2+28uz+22u+260z^2+1860z+1703$. Hence $Z_0 > 2^R$. The signed Pell identities modulo the odd H give either $X2^R = \pm 1 \pmod H$ or $X = e2^R \pmod H$, $e \in \{\pm 1\}$. The strict bounds $1 < |X|2^R < |H|-1$ and $|X - e2^R| < |H|$ exclude the reciprocal branch and unwrap the direct one. Therefore $X = e2^R$ with odd R. Only now does $X \equiv 2 \pmod 3$ force $e = 1$. Restoring $u_{\text{old}} = 2^{(R-1)/2}$ recovers the missing row $X = 2u_{\text{old}}^2$ and every original Q41 equation. The inherited Q41/native/calibration theorem supplies the finite refutation.

3.3 Why completeness survives both restrictions

The canonical source-34 witnesses have $q = 2^t$ with odd t, $R \equiv 3 \pmod 4$, and $X = 2^R$; thus $(X+1)/3$ is integral. The new determinant holds for this same canonical family, including all permitted rank-controller dummy changes. In radix $\rho = 2^{121}$, each occupied cell has its fixed top C digit at position 3296 and its fixed top F digit at position 8791, of coefficient $c_0 = 105534389824824194647382298625$. The carry-free native bounds and literal scale identities

$$121 \cdot 8791 + 1113 = 1064921 - 97, \qquad 121 \cdot 3296 + 666104 = 1064921 - 1$$

imply, uniformly in tableau size and rank tuning,

$$q/2 \leq U < 4q/5, \qquad q/2 \leq V_0 < 3q/5.$$

Writing $q = 2s$, $U = s+x$, $V_0 = s+y$ gives $0 \leq x < 3s/5$, $0 \leq y < s/5$, so

$$UV_0 - (q-U-V_0)^2 = s^2 + s(x+y) - x^2 - xy - y^2 > \frac{12}{25}s^2 > 0.$$

A Pos representation supplies the three new guard coordinates. Thus the same canonical witnesses satisfy both restrictions simultaneously. This does not assert that every old zero satisfies the new determinant.

3.4 The sextic and octic companions

For projections use the equally valid affine quartic $J_{\text{old}} = 3J_3 + 1$, $w_{\text{old}} = 3w_3 + 1$ rather than replacing X. It still has $X \equiv 2 \pmod 3$. Canonical completeness holds because $\beta \equiv 1 \pmod 3$ gives $J_{\text{old}} \equiv 1 \pmod 3$, while $w_{\text{old}} = 2^{R-t-2} \equiv 1 \pmod 3$ has a nonnegative even exponent. Put $q = \beta(3J_3 + 1) + 1$ and $W = 4(3w_3 + 1)$.

For the sextic set $\tau_{\text{old}} = \tau_{\text{tail}} + E(c - \eta)$ and restore

$$P = k + E(c - \eta), \quad N = q(L + 1), \quad X = Wq, \quad C_2 = c^2, \quad F_2 = f^2, \quad g = y^2 - V^2.$$

Six coordinates and six defining rows disappear. The first norm becomes $\tau_{\text{tail}}^2 + (2\tau_{\text{tail}} - k)E(c - \eta) - 1$. All 18 residuals in 31 variables have degree at most three, and their sum of squares has coefficient four at r^2c^4.

For the octic put $k_0 = 1 - V + jc + hE$ and restore

$$L = q^2 - 1, \quad N = q^3, \quad X = Wq, \quad C_2 = c^2, \quad F_2 = f^2,$$
$$g = y^2 - V^2, \quad B_{\text{aux}} = I^2, \quad D = Wq + ac + \gamma(4a + 3), \quad \Delta = a^2 + 4a + 3,$$
$$\zeta = k_0 - \eta, \qquad P = k_0 + E(c - \eta).$$

Eleven coordinates and eleven rows disappear. The 13 residuals in 26 variables have degree at most four, and the sum of squares has coefficient one at a^4f^4. These are exact integer-polynomial zero-set bijections, with no hidden division. Explicit restorations are contained in each JSON.

Alternative and checks. Source 69's weighted-cone guard also composes, giving the same three sizes. Its canonical fixed-anchor bounds survive the same parity tuning, but its six numerals require 881,696 raw bytes, so the determinant is preferred. The checker verifies all eight packets against source 70's literal packets, all restoration identities and authentic coefficient payloads, liveness, exact degrees, and altered-guard negative controls. Finite replay is evidence for those algebraic and fixed-data claims, not a substitute for the unbounded mathematical arguments or a new formal proof of the repository's internal consistency biconditional.

4 A complete compact equation in twenty-four integers

This section prints a common base for the smaller projections, so that the construction is more than a table of claimed sizes. Take the six fixed integers $\beta, K_0, M_C, M_F, A_0, B_0$ bound in the preceding section, with $A_0 = 2^{1113}$ and $B_0 = 2^{666104}$. The 24 unrestricted unknowns are

$$J_3, F, \alpha, z, f, h, r, v, w_3, \tau, \eta, y, \gamma, o, j, E, c, a, V, I, t, p_1, p_2, p_3.$$

The aliases v, t, p_i stand for `v_scale`, `pair_root`, and `pair_pos`i in the literal JSON. Define

$$J = 3J_3 + 1, \qquad q = \beta J + 1, \qquad W = 4(3w_3 + 1),$$
$$X = qW, \qquad Y = (2v + 1)q^3, \qquad C = q - F - \alpha,$$
$$k = 1 - V + jc + hE, \qquad H = 4a + 3, \qquad \Delta = a^2 + H,$$
$$R = [(q - 1)(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J,$$
$$D_0 = (A_0F)(B_0C) - (q - A_0F - B_0C)^2, \qquad T_0 = \eta(k - \eta),$$
$$\Pi = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).$$

Every displayed symbol on a left-hand side is only an abbreviation. The twelve residuals are

$$\begin{aligned}
e_1 &= a - E - Y,\\
e_2 &= E(X+1) - aX,\\
e_3 &= c - k(a-E) - \eta,\\
e_4 &= \tau^2 - E(c-\eta)\big[k + E(c-\eta)\big] - 1,\\
e_5 &= (X + ac + \gamma H)^2 - \Delta c^2 - 1,\\
e_6 &= jc - V - R,\\
e_7 &= (K_0 + W)C - F - z(q-1),\\
e_8 &= V + c - fo,\\
e_9 &= I - (2r+1)c^2,\\
e_{10} &= I^2 - \Delta(f^2 - 1),\\
e_{11} &= I^2V^2 - (I^2-1)y^2 - 1,\\
e_{12} &= D_0T_0 - (t^2 - D_0 - T_0 + 1)^2 - \Pi.
\end{aligned}$$

Then

$$\boxed{P_{24} = e_1^2 + e_2^2 + \cdots + e_{12}^2 = 0, \qquad \boldsymbol{x} \in \mathbb{Z}^{24}.}$$

It has exact degree 12. The new last row forces $D_0 > 0$ and $T_0 > 0$ by the universal simultaneous-positivity lemma below. It replaces two separate ternary positivity witnesses by four witnesses, without changing the other arithmetic coordinates.

For the 25-variable octic, add an unrestricted ζ, add the residual $\zeta - k + \eta$, and use $T_0 = \eta\zeta$ only in the last row. Keep k and all other rows exactly as printed. This is an existential extension whose projection is the 24-variable equation, but its degree is8 because the last row is now quartic rather than sextic. The coefficient of t^8 is 1. The independent root script `check_displayed_system.py` expands and compares every printed row with both literal JSON artifacts.

The pair lemma and the further projections are proved next. In particular, when a row is replaced using another retained equation, we assert equality of zero sets, not equality of the entire sum-of-squares polynomials.

5 Joint positivity and a projection family

All variables in this section range over unrestricted integers. The six fixed numerals are the unchanged authentic source-34 tuple and the two determinant scales. Write $d = U_cV_c - (q - U_c - V_c)^2$, where $U_c = 2^{1113}F$ and $V_c = 2^{666104}(q - F - \alpha)$. The separate symbol $W = 4(3w_3 + 1)$ denotes the scale in the compact arithmetic system. These are polynomial abbreviations, not additional unknowns.

5.1 An elementary universal two-sign lemma

The following construction is the joint-positivity lemma of source 68, *Joint Positivity and Free Lucas*, Sections 1–2 of its `PAIR_GUARD_PROOF.md`. We claim its composition with the new determinant/residue system, not the lemma as a new discovery.

Put

$$G(x,y,t) = xy - (t^2 - x - y + 1)^2, \qquad \mathrm{Pos}(p_1,p_2,p_3) = 1 + p_1^2 + p_2^2 + p_3(p_3+1).$$

For integers x, y,

$$x > 0 \;\&\; y > 0 \quad\Longleftrightarrow\quad \exists t, p_1, p_2, p_3 \in \mathbb{Z} : G(x,y,t) = \mathrm{Pos}(p_1,p_2,p_3). \tag{JP}$$

Indeed, $G > 0$ implies $xy > 0$. If $x = -a, y = -b$ with $a, b \geq 1$, then $(t^2 + a + b + 1)^2 \geq (a + b + 1)^2 > ab$, a contradiction. This proves soundness for every signed assignment, including arbitrary t.

Conversely, for $x, y \geq 1$ set $N = x + y - 1$ and $m = \lfloor \sqrt{N} \rfloor$. Choose $t = m$ when $N - m^2 \leq m$, and $t = m + 1$ otherwise. Then $|t^2 - N| \leq m$. The error is zero when N is a square; otherwise $(t^2 - N)^2 \leq m^2 < N \leq xy$, since $xy - N = (x - 1)(y - 1) \geq 0$. Thus $G > 0$. Every positive integer n is a value of Pos: apply the three-square theorem to $4n - 3$, which is positive and congruent to 1 or 5 modulo 8. Exactly one square is odd, so the representation has the form $(2p_3 + 1)^2 + (2p_1)^2 + (2p_2)^2$. Division by four gives the asserted identity. The floor and square root are only a witness-selection recipe in this proof; neither appears in a polynomial.

5.2 The 25-variable octic and its 24-variable projection

The 26-variable determinant/residue octic has eleven arithmetic rows and two private ternary positivity rows asserting

$$d > 0, \qquad \eta\zeta_0 > 0, \qquad \zeta_0 = 1 - V + jc + hE - \eta.$$

Delete those six private coordinates and two rows. Appending

$$d\eta\zeta_0 - (t^2 - d - \eta\zeta_0 + 1)^2 - \mathrm{Pos}(p_1, p_2, p_3) = 0 \tag{J24}$$

gives 24 variables and 12 rows. Its joint row has degree six. Its top part is $-\eta^2(hE + jc)^2$, so the sum of squares has exact degree twelve, with coefficient $+1$ at $\eta^4 h^4 E^4$.

To keep degree eight, retain one extra coordinate ζ, impose $\zeta = \zeta_0$, and replace ζ_0 by ζ only in the joint row. Both d and $\eta\zeta$ are now quadratic, so the new row has degree four. This gives 25 variables and 13 rows. Its coefficient at t^4 is -1; no other row mentions t. Hence the final polynomial has coefficient $+1$ at t^8 and exact degree eight. The eleven old arithmetic rows are unchanged and may still contain ζ_0; the restoring row makes this intentional choice equivalent to using ζ everywhere.

For either construction, (JP) restores the two old positive quantities, and their ternary representations restore all six deleted coordinates. Conversely, every parent zero supplies those quantities and therefore the four new witnesses by the explicit recipe above. Thus this step preserves existence at every fixed assignment of retained arithmetic coordinates. It requires no additional canonical restriction. Canonical completeness is inherited only where it is actually used upstream, in the determinant and residue constructions.

5.3 A degree-sensitive row operation

Use the compact system's abbreviations

$$X = 4q(3w_3 + 1), \quad Y = (2v_{\mathrm{scale}} + 1)q^3, \quad k = 1 - V + jc + hE, \quad \Delta = a^2 + 4a + 3.$$

The relevant rows are

$$a - E - Y = 0, \quad E - X(a - E) = 0, \quad c - \eta - k(a - E) = 0,$$
$$V + c - fo = 0, \quad I - (2r + 1)c^2 = 0,$$
$$S = I^2 - \Delta(f^2 - 1) = 0, \quad A = I^2(V^2 - y^2) + y^2 - 1 = 0.$$

Define $A_{\mathrm{red}} = \Delta(f^2 - 1)(V^2 - y^2) + y^2 - 1$. The exact identity

$$A - A_{\mathrm{red}} = (V^2 - y^2)S \tag{RI}$$

allows either auxiliary row while S is retained. It is a two-sided zero-set row operation, not equality of the two free polynomials.

Every elimination below uses a monic defining row and an integer-polynomial inverse. Substitution maps old zeros to new zeros; restoring the deleted coordinate makes its defining row zero and recovers every other old row. There is no division, sign assumption, or new Pell existence theorem.

- At 23 variables, eliminate $V = fo - c$ and keep A. The auxiliary and joint rows have degree six, so the final degree is twelve. Alternatively, use (RI), eliminate $I = (2r+1)c^2$, and keep V; the degree is again twelve. The latter coordinate choice has a smaller expanded support in the exact independent audit, but no claim of circuit minimality follows.
- At 22 variables, use (RI) and eliminate both V, I. The reduced auxiliary has degree eight and top term $a^2f^4o^2$, giving final degree sixteen.
- At 21 variables, instead retain the original A and eliminate V, I, a, with $a = E + Y$. The main, strong, and auxiliary rows have degree at most ten; all other rows have degree at most six. This gives degree twenty.
- At 20 variables, still keep the original A and retain η. The first two rows give $E = XY$ and $a = (X+1)Y$. Together with the V, I restorations, these leave eight rows. The joint row has degree sixteen, the main/strong and first Pell rows have degree at most fourteen, and the original auxiliary has degree ten. The final degree is thirty-two.
- At 19 variables, also eliminate $\eta = c - Yk$. The joint row has degree forty-four and the other six rows have degree at most thirty-four. The final degree is eighty-eight.

The choice of auxiliary row is essential for the degree ledger. Keeping A_{red} at the 20-variable endpoint would needlessly raise its degree. These endpoints may all be defined directly from the compact 24-variable source by the stated substitutions and, only where specified, the row identity (RI).

5.4 Exact-degree certificates on the authentic slice

For brevity in this table only, write $J = J_3$, $v = v_{\mathrm{scale}}$ and $w = w_3$. Each entry is an exact coefficient of the final sum of squares.

Variables	Degree	Monomial	Coefficient
25	8	t^8	1
24	12	$\eta^4h^4E^4$	1
23	12	$\eta^4h^4E^4$	1
22	16	$a^4f^8o^4$	1
21	20	$f^4v^4J^{12}$	$2^4 3^{12}\beta^{12}$
20	32	$\eta^4h^4w^4v^4J^{16}$	$2^{12}3^{20}\beta^{16}$
19	88	$h^8w^8v^{16}J^{56}$	$2^{32}3^{64}\beta^{56}$

All are positive because the authentic fixed $\beta = 2^{1064921} - 1$ is positive. The 23-variable certificate applies to both coordinate choices.

Here is an unbounded check of the last three entries. The leading forms are

$$X_* = 36\beta Jw, \qquad Y_* = 54\beta^3 J^3 v.$$

At 21 variables the strong row has top term $-Y_*^2 f^2$. The main row has no f, and the original auxiliary has no J, so the displayed coefficient can come only from the strong square. At 20 variables, $E_* = X_*Y_* = 1944\beta^4 J^4 wv$ and the joint top form is $-\eta^2 h^2 E_*^2$; its square is the only contribution of degree 32. At 19 variables,

$$\eta_* = -hX_*Y_*^2 = -2^4 3^8 \beta^7 h J^7 w v^2.$$

Since $\deg \eta > \deg k$, the joint top form is $-\eta_*^4$. Its square gives the displayed degree-88 monomial, with no other residual of sufficiently high degree to contribute. These arguments preclude cancellation at the actual numeral values, rather than merely asserting a generic gate-degree bound.

The 19-variable coordinate list is exactly

$$J_3, F, \alpha, z, f, h, r, v_{\mathrm{scale}}, w_3, \tau, y, \gamma, o, j, c, t, p_1, p_2, p_3.$$

In particular X, Y, E, a, V, I, η and the coefficient ports are not hidden quantified variables.

5.5 Certificate scope

All residuals are serialized as ordinary, completely inlined polynomial expressions. High-degree packets retain factorization to avoid a wasteful expanded sum of squares. A separate standard-library checker authenticates the compact source identities, the row operation, every deleted defining row, the literal surviving-row substitutions, the declared variable set, and the exact homogeneous coefficient certificates. It passes both normal and optimized Python execution. Finite tests are not substituted for the universal proof of (JP), the inverse maps, or the inherited compiler, Pell, and refutation-calibration theorems.

Consequently the ordinary mathematical existence equivalence with a finite ZFC refutation extends to every listed packet. No new kernel-checked Lean/Rocq theorem or encoded object-ZFC derivation is claimed, and this bounded family of variable/degree tradeoffs is not an optimality theorem.

6 Literal shared circuits for the integer family

The residual systems also admit small shared binary evaluators. The fixed numeral leaves are free in the same counting convention as the input packages; every live scalar multiplication, squaring, addition and subtraction is charged. Registers are reused, not re-expanded into expression trees. The following counts specify delivered circuits, not optimal circuits.

Integer unknowns	Exact degree	One polynomial	Residual system
25	8	127	102
23	12	121	100
22	16	118	99
21	20	115	98
20	32	112	97
19	88	109	96

The system column charges the union of live gates needed to evaluate all residuals, not the sum of separately recomputed expressions. The polynomial column additionally forms and sums their squares. The 23-variable circuit here is the variant eliminating V; the distinct variant eliminating I has a smaller expanded support and is supplied separately.

The circuit proof is literal polynomial equality, stronger than equality of zero sets. It validates each restored-coordinate definition and each residual with named symbolic intermediate expressions, then structurally inlines those expressions to match the corresponding JSON packet. For example, the nineteen-variable first norm simplifies using the exact small-ring identity

$$(XY)(kY)\big(k + (XY)(kY)\big) = XY^2(XY^2 + 1)k^2.$$

Such identities avoid expanding the entire degree-88 sum of squares while still proving an identity for every integer assignment. Random modular evaluation alone would not suffice. The independent checker separately checks topology, liveness, all declared inputs, counted binary operations, and the symbolic residual bindings. No modular test is substituted for an unbounded circuit identity.

The degree-88 equation in 19 unrestricted integers thus uses 109 operations, with the same six numerical payloads as the other determinant constructions. That improves the selected older 24-integer degree-1936/161-operation point on variables, degree and this circuit-cost convention. Coefficient payloads and proof dependencies are still separate comparison axes, so this is not a global claim of the simplest possible equation.

7 Denominator-free and centered three-root transport

Two new eleven-positive-variable endpoints improve the inspected degree-325, 113-operation construction of source 70:

$$\boxed{\deg P_{11} = 277, \quad 111 \text{ operations}}, \qquad \boxed{\deg \widetilde{P}_{11} = 257, \quad 114 \text{ operations}}.$$

The first deletes a denominator guard using the adjacent-coefficient Pell descent of source 69. The second additionally centers the auxiliary root and proves transport positivity by a new signed bootstrap. They are different degree/cost choices. Every multiplication by a numeral and every square is charged; fixed numeral leaves and register reuse are free. Neither construction changes the known nine-positive-variable bound or claims global optimality. All native compiler and literal machine/refutation calibrations remain named inherited dependencies.

7.1 A denominator guard that can be deleted

The following elementary fact replaces the prime-support argument in source 70.

Adjacent-coefficient descent. If $Q > 1$ and V are integers, y is rational, and

$$QV^2 - (Q-1)y^2 = 1,$$

then y is an integer. Indeed, replace both coordinates by their absolute values and write $y = a/m$ in lowest terms. Then $m^2 \mid Q - 1$. For $V > 1$ apply

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV.$$

Here V' is integral, the same norm is preserved, and $y > V$ and $QV > (Q-1)y$ give $-V < V' < V$. The norm excludes $V' = 0$. Since $2Q - 1 \equiv 1 \pmod{m}$, the reduced denominator of y' is still m. Strong induction reaches $V = 1$, where $y = 1$, and proves $m = 1$.

Thus, after a sum-of-three-roots argument has made the auxiliary root rational, no denominator constraint involving the first transport modulus is needed. In particular source 70's $I = iq(q-1)c^2$ can be replaced by $I = ic^2$, deleting two multiplication gates. The unconditional separation proof used only $I \geq c^2$ and remains valid.

7.2 Complete polynomials

The eleven unknowns are

$$j, F, \alpha, i, T, X, Y, n, \eta, \zeta, \gamma \in \mathbb{Z}_{>0}.$$

The four fixed numeral ports are the complete compact44 tuple β, K_0, M_C, M_F, with $\beta = 2^{590069} - 1$ and M_F already shifted. They are not free parameters; their authenticated payloads and whole-tuple binding accompany the circuits. Every other symbol below is a polynomial register, not another quantified

variable. Define

$$J = 3j + 1, \qquad q = \beta J + 1, \qquad C = q - F - \alpha,$$
$$E = XY, \qquad k = \eta + \zeta, \qquad c = kY + \eta,$$
$$a = E + Y, \qquad H = 4a + 3, \qquad \Delta = a^2 + H = (a+2)^2 - 1,$$
$$\ell = EkY, \qquad I = ic^2, \qquad M = q^3(q-1),$$
$$R = ((q-1)(q-F) + \alpha)(q^2-1) + (M_C + qM_F)J,$$
$$\widehat{R} = q^3(q + F + \alpha), \qquad B_t = (qK_0 + X)C - qF,$$
$$D_\tau = \ell(\ell + k) + 1, \qquad D_f = 1 + \Delta I^2,$$
$$Q = \Delta^2 I^2, \qquad \delta = Q - 1, \qquad V = D_f(cT - R) - c,$$
$$r_0 = k - R - 1, \qquad S = 4qY, \qquad U = qS,$$
$$D_* = U(X + ac) + H(U\gamma + SX - r_0),$$
$$A_1 = D_*^2 - U^2(1 + \Delta c^2).$$

For the degree-277 endpoint choose

$$L = 3D_f(cT + \widehat{R}), \quad h_* = L, \quad W = 1 + (q-1)h_*, \quad Z = M(\ell + k + n + L) + (q-1)Y - q^3 W B_t.$$

For the degree-257 endpoint instead choose

$$B_0 = cT + \widehat{R}, \qquad L = D_f + B_0,$$
$$h_* = 4qD_f + L, \qquad W = 1 + (q-1)h_*,$$
$$Z = M(\ell + k + n + L + V) + (q-1)Y - q^3 W B_t.$$

The shared division-free final norm is

$$A = M^2 D_\tau, \qquad B_f = M^2 D_f, \qquad C_y = M^2(QV^2 - 1),$$
$$T_N = \delta^2 A B_f, \qquad R_N = \delta C_y,$$
$$J_N = \delta(Z^2 - A - B_f) - C_y,$$
$$N_c = J_N^2 - 4\big(T_N + R_N(A + B_f)\big),$$
$$N_8 = N_c^2 - 64 T_N R_N Z^2,$$
$$P = N_8 + A_1 + N_8 A_1.$$

The literal graphs are `p11_positive_degree277.json` and `p11_positive_degree257.json`.

7.3 Three-root separation and the degree-277 lift

In proof only put $D_y = (QV^2 - 1)/(Q - 1)$, and let $t = \sqrt{D_\tau}$, $f = \sqrt{D_f}$ and $y = \sqrt{D_y}$ be positive. For every positive input, independently of any equation or sign of C,

$$1 < D_\tau < D_f < Q, \qquad |V| > 1, \qquad D_y > V^2.$$

The radicand D_y cannot equal either integer $D \in \{D_\tau, D_f\}$: that would give $Q(D - V^2) = D - 1$ with $0 < D - 1 < Q$. There is also strict root dominance. Write $b = cT - R$. If $b = 0$, then $V = -c$, $y < c + 1$ and $f > t + y$. If $b \neq 0$, then $|V| \geq D_f - c > 2f > f + t$, hence $y > f + t$. These estimates use $I \geq c^2$, $c \geq 3$ and $f > (a+1)I \geq 3c^2$. Thus no signed sum of the three roots is zero.

The universal octic sum norm is

$$F_8(x; a, b, c) = \big[(x^2 - a - b - c)^2 - 4(ab + ac + bc)\big]^2 - 64abcx^2.$$

It is the product of the eight signed-root factors. The displayed N_8 is exactly $\delta^4 F_8(Z; A, B_f, C_y/\delta)$. At $N_8 = 0$, therefore,

$$Z/M = \tau_* + f_* + y_*, \qquad \tau_*^2 = D_\tau, \quad f_*^2 = D_f, \quad y_*^2 = D_y.$$

A rational signed sum of three distinct square roots, with no zero signed sum, forces each root rational. One exact extraction is as follows: for signed roots x_1, x_2, x_3 with rational nonzero sum s and squares a, b, c, put

$$u = (s^2 - a - b - c)/2, \quad v = (u^2 - ab - ac - bc)/(2s), \quad x_1 = \frac{a(a+u)}{as+v}.$$

The denominator is $x_1(x_1 + x_2)(x_1 + x_3) \neq 0$; cycle the formula. Consequently τ_*, f_* are integers by rational-square integrality, and y_* is an integer by the preceding Pell descent.

For the 277-degree construction, $L > f + y$ unconditionally. Recover

$$r = n + L - f_* - y_* > 0, \quad f_{\text{old}} = |f_*|, \quad i_{\text{old}} = i, \quad u = |y_*| - V > 0.$$

These are precisely the weighted source equations. Conversely choose canonical strong and auxiliary source witnesses first, then L, W and the transport witness. The inherited weighted theorem gives $r > qW - 2k > L$, so $n = r - L + f + y > 0$. No stronger divisibility than $c^2 \mid I$ is required.

7.4 The new centered sign bootstrap

The centered endpoint cannot assert $r > 0$ before proving $V > 0$. The following order resolves that issue.

First recover all three integral roots as above. Since $M \mid Z$, reduction modulo q^3 and $q - 1$ gives

$$q^3 \mid Y, \qquad q - 1 \mid B_t.$$

No transport positivity is used here. Also $A_1 = 0$ makes D_*/U integral; classify its *absolute value* as a positive Pell root. Together with the first Pell root, this gives

$$c = \psi_{a+2}(p), \quad k = 2\psi_{2XY^2+1}(m), \quad p, m \geq 1.$$

Here $c \geq 3$ implies $p \geq 2$. The case $m = 1$ would give $k = 2$ and $c = 2Y + 1 < 2(a + 2)$, a contradiction. Hence

$$c > 8XY^3 \geq 8q^9. \tag{*}$$

The authentic mask bounds give $|R| < \widehat{R}$, hence $|V| < D_f B_0$, while $\delta > D_f$ and $D_f > f$. Therefore

$$0 < y - |V| < |V|/\delta < B_0.$$

In particular $L + V > f + y$ if $V > 0$.

Suppose now that $C \leq 0$. Set $d = -C \geq 0$ and $B = -B_t = (qK_0 + X)d + qF > 0$. Since $K_0 \geq 1$ and $(M_C + qM_F)J < 2q^2$, direct expansion gives

$$qB \geq q^2 d + q^2 F, \qquad R < qB + q^4 + q^2.$$

Writing $s = Y/q^3 > 0$, the root equality becomes

$$\tau_* + f_* + y_* = \ell + k + n + L + V + s + \frac{WB}{q-1}. \tag{**}$$

If $V > 0$, the right side is greater than $t + f + y$, impossible. If $V < 0$, then the integer $cT - R$ is nonpositive, so $R \geq cT \geq c > 8q^9$. Thus $q^4 + q^2 < R/2$ and $R < 2qB$. Moreover $|V| < D_f R$, so $L + V - f - y > -2D_f R$, whereas

$$\frac{WB}{q-1} > 4qD_f B > 2D_f R.$$

Again the right side of $(**)$ exceeds $t + f + y$. The impossible case $V = 0$ was already excluded. This proves $C > 0$.

Only now use the authentic bound $3q < R < q^4 - 2$. Together with $(*)$ it gives $V > 1$, and therefore

$$r = n + L + V - f_* - y_* \geq n + L + V - f - y > n > 0.$$

The old positive strong and auxiliary coordinates are recovered exactly as before. The weighted carrier soundness theorem now applies.

For completeness choose canonical strong and auxiliary data before computing L, h_*, W. Since $h_* > L + k$, the canonical transport variable satisfies $r > qW - 2k > L$. Set

$$n = r - L - V + f + y = r - L + f + u > 0.$$

This gives the displayed centered norm. The old positive main-carrier coordinate γ is unchanged. This is an existence equivalence, not a bijection between every old and new zero set.

Both N_8 and A_1 are squares modulo four. Thus $(1 + N_8)(1 + A_1) = 1$ forces both factors to be 1, since -1 would require a residual -2. The single equation is therefore equivalent to the two residual equations. Under the inherited interfaces, positive solvability is equivalent to a finite ZFC refutation; negation gives the stated consistency equivalence.

7.5 Exact degrees and checks

For the uncentered construction the exact degrees of $I, D_f, Q, V, L, W, Z, N_c, N_8, A_1$ are respectively

$$5, 14, 18, 18, 18, 19, 24, 132, 264, 13.$$

The coefficient of $j^{60}i^{24}\eta^{49}X^{42}Y^{85}\alpha^{16}\gamma$ in the degree-277 polynomial is $2^7 3^8 (3\beta)^{60} > 0$.

For the centered construction put $z = Z - MV$ in proof only. The exact cancellation identity is

$$J_N = \delta\{z(z + 2MV) - A - B_f\} - M^2(V^2 - 1).$$

Here $\deg z = 21$, $\deg Z = 22$, and the five displayed summand degrees are $60, 61, 34, 40, 44$. Thus $\deg J_N = 61$, $\deg N_c = 122$ and $\deg N_8 = 244$; the subtracted norm correction and tail have degrees 102 and 198, respectively. The coefficient of

$$j^{52}i^{24}\eta^{49}X^{38}Y^{85}F^4\alpha^4\gamma$$

is $2^{19}(3\beta)^{52} > 0$, proving exact degree 257 after actual fixed-port specialization.

Both standard-library builders reproduce the literal graphs byte for byte. Both exact checkers pass normally and under `python -O`: prefix identities, the eight-factor norm, rational extraction identities, main-carrier cancellation, centered cancellation, degree heads, selected coefficients, gate ledgers and fixed coefficient hashes. Independent review receipts are kept separately. These finite checks do not prove the unbounded native/compiler interfaces or the consistency of ZFC.

Unfinished four-root direction. A direct ten-variable extension is not established. Distinct radicands and a nonzero full signed sum no longer ensure rationality root by root: $\sqrt{2} + \sqrt{8} - \sqrt{18} + 100 = 100$, with four distinct radicands and no vanishing full signed sum. Even a stronger separation argument would not remove the main-residue versus transport-modulus obstruction from the earlier attempted merger. The negative controls in this component record both issues; no ten-variable endpoint is claimed.

8 Smaller local graphs for the direct checker

The latest pinned repository supplies a complete numeric checker for the original sealed-ZFC calculus, independent of K215 and Jones finalization. Its 83,281 raw and 76,313 optimized primitive-recursive program nodes are not polynomial variable counts. This section improves concrete local polynomial graphs used by that route, without changing its numeric codes. Every visible variable and witness below ranges over $\mathbb{N}$, including zero; every subtraction is ordinary integer-polynomial subtraction.

Local graph	Old witnesses/ops	New witnesses/ops	Degree
Max-square pair	3/46	2/39	4
Pair, strict squared rows	3/46	2/42	4
Either unpair projection	4/46	3/39	4
Predecessor	0/16	0/10	4
Tree left/right, list tail	5/62	4/49	4
If-zero conditional	2/32	1/16	4
Zero test	1/20	1/14	4
Actual list head	7/76	5/59	4
Pair, selector eliminated	3/46 (quartic)	1/53	6

Operations count binary additions, subtractions and multiplications in the literal expression tree, including the final addition of zero, as in the baseline. Repeated subtrees are counted repeatedly. The artifacts also provide separate shared-DAG counts and full expanded integer coefficients. No local saving is advertised as a completed smaller consistency polynomial.

8.1 Remove the maximum-component witness

For natural z, a, b, t, s, put

$$B = t(t-1), \qquad L = a - b - s + t(2s+1), \qquad E = 4(z-a-b) - (a+b+s)^2 + t,$$

$$\mathrm{Pair}(z, a, b; t, s) = B + L^2 + E^2.$$

Then

$$\exists t, s \in \mathbb{N} : \mathrm{Pair} = 0 \quad \Longleftrightarrow \quad z = \begin{cases} b^2 + a, & a < b, \\ a^2 + a + b, & b \le a. \end{cases}$$

Indeed $t(t-1) \ge 0$ for every integer t, so a zero forces $t = 0$ or $t = 1$ and $L = E = 0$. At $t = 0$, $a = b + s$ and $a + b + s = 2a$; hence $z = a^2 + a + b$. At $t = 1$, $b = a + s + 1$ and $a + b + s = 2b - 1$; hence $z = b^2 + a$. Conversely these two branches choose respectively $(t, s) = (0, a - b)$ and $(1, b - a - 1)$. This covers ties and zero. The literal quartic has two witnesses, 39 tree operations and 52 monomials. Replacing B by B^2 gives a strict sum of three squared quadratic rows, with 42 operations and the same zero set. This latter version fits the existing squared-row interface without a nonnegative-term extension. Hiding the missing component yields either unpair projection with three witnesses.

Solving the last row for the selector also gives the one-witness sextic

$$T = (a+b+s)^2 - 4(z-a-b), \qquad T(T-1) + \big(a - b - s + T(2s+1)\big)^2 = 0.$$

Here T is an integer polynomial, not a new coordinate. Its nonnegative first summand forces $T \in \{0, 1\}$; the preceding branch proof then applies. This has 53 tree operations and 86 monomials, but raises degree to six.

8.2 Share the predecessor and remove branch selectors

The exact predecessor graph factors as

$$[n(p+1-n)]^2 + [p(p+1-n)]^2 = (n-p-1)^2(n^2+p^2).$$

Its zero set is precisely $p = \max(n-1, 0)$: either $n = p+1$ or $n = p = 0$. Adding this product to $\mathrm{Pair}(p, a, b; t, s)$ gives either tree projection with four hidden coordinates and 49 operations. At $n = 0$ both components are zero, so the right projection is also the actual guarded list tail.

A separate selector is unnecessary for a conditional. The quartic

$$[n(c-b)]^2 + [(c-a)(n-s-1)]^2 = 0$$

with one natural witness s is equivalent to $c = a$ if $n = 0$, and $c = b$ otherwise. At zero, $n - s - 1 = -s - 1 \neq 0$; at positive n, the first row forces $c = b$, and completeness chooses $s = n - 1$. Taking $a = 1, b = 0$ gives the 14-operation zero test. If an exact predecessor p is already available, set $s = p$ and allocate no extra slack.

For the actual found-head convention, with visible n, c and witnesses p, a, b, t, s, use

$$\mathrm{Head} = \mathrm{Pair}(p, a, b; t, s) + (n-p-1)^2(n^2+p^2+c^2) + [n(c-a-1)]^2.$$

At $n = 0$, its middle term forces $p = c = 0$ and the pairing term forces $a = b = 0$. At $n > 0$, it forces $p = n - 1$, $c = a + 1$, and the correct unpair components. Conversely these choices make every term zero. Thus nil is zero and a found zero-valued entry is one. The ordinary quartic has five witnesses, 59 operations and 72 monomials.

8.3 Certificate status

The package supplies literal trees and full coefficient lists for eleven local polynomials. Exact symbolic identities, counts and 1,845,448 finite controls pass in ordinary and optimized Python modes; checks remain active under `-O`. Universal correctness follows from the arguments above, not from the finite controls. These new proofs have not been checked by Lean, and the whole numeric checker, its bounded loops and its internal nonstandard-code interpretation still require integration. Natural witness domains are essential; unrestricted integer witnesses need their own naturalness encoding.

9 Index-free inert-prime closure with a reused product scale

Fix $m \geq 1$, natural n, external natural parameters x, and fixed nonnegative-coefficient polynomials $e(i, w; x), f(i, w; x)$. Set $P = e - f$ and choose a fixed $d \geq 0$ bounding its degree in i; for $P = 0$, $d = 0$ is allowed. For a natural witness bound U, define

$$\begin{aligned} H &= e(n, U, \ldots, U; x) + f(n, U, \ldots, U; x) + n + U + 1, \\ T &= H!, \qquad D = 4T, \qquad S = D^n n!, \qquad M = DS + 1, \\ B &= \left(S\tbinom{S+n}{n}\right) \bmod M. \end{aligned}$$

Writing $P(i, w; x) = \sum_{k=0}^{d} P_k(w; x) i^k$, set

$$R_D(A; x) = \sum_{k=0}^{d} P_k(A; x) D^{d-k} (1-D)^k.$$

This is an ordinary polynomial. Its inverse-looking shorthand $D^dP(D^{-1}-1,A;x)$ introduces no actual inverse or variable exponent. For a fixed nested binary norm N, obtained by repeatedly replacing a pair of subexpressions by the sum of their squares, the exact certificate is

$$\exists U, A_1, \ldots, A_m \in \mathbb{N}: \quad B \mid N\left(\tbinom{A_1}{U+1}, \ldots, \tbinom{A_m}{U+1}, R_D(A;x)\right).$$

It is equivalent to $\forall i < n\ \exists w \in \mathbb{N}^m : P(i,w;x) = 0$. Binomial is total, with $\binom{A}{k} = 0$ for $A < k$. Neither factorials nor binomials nor D^n are being called ordinary polynomial notation.

Why the product needs only one variable-power graph. For $m_i = D(i+1)-1$,

$$S\binom{S+n}{n} = \prod_{j=1}^{n}(DS+Dj) \equiv \prod_{j=1}^{n}(Dj-1) \pmod{DS+1}.$$

For $n > 0$ the positive product is smaller than $D^n n! = S < M$; at $n = 0$ it is one, still below M. Hence $B = \prod_{i<n} m_i$. The former independent bound $(Dn)^n + 1$ is unnecessary: reuse S itself. This removes one complete variable-power graph, not merely its displayed abbreviation.

Why the encoded index is unnecessary. The integer polynomial $R_D - D^dP(i,A;x)$ is divisible by $D(i+1)-1$, by the difference-of-powers identity. Thus modulo m_i, the cleared residual is exactly $D^dP(i,A;x)$. The modulus already encodes the index as $D^{-1}-1$, so no CRT coordinate c or index norm leaf is needed.

Each m_i is 3 modulo 4, all its prime factors exceed H, and the m_i are pairwise coprime: a common divisor divides $j-i$, hence $H!$, but is coprime to D. Choose a prime $p_i \equiv 3 \pmod 4$ dividing m_i. Anisotropy of a binary sum of squares over $\mathbb{F}_{p_i}$ makes every norm leaf zero. The binomial leaves recover residues $w_j \in [0,U]$; D is invertible, so the last leaf gives $P(i,w;x) = 0 \pmod{p_i}$. Its original absolute value is smaller than $H < p_i$, hence it is zero as an integer. The bound is applied after cancelling D^d, not to the large cleared residual. Conversely CRT representatives of actual witnesses make every leaf divisible by every m_i, proving completeness. The empty bound $n = 0$ is handled by $B = 1$.

An exposed, conditional witness ledger. Suppose chosen factorial, power and total-binomial graph polynomials use respectively ν_F, ν_P, ν_B internal witnesses beyond visible arguments and output. Expose graph outputs

$$T = H!, \quad F = n!, \quad E = D^n, \quad C = \binom{EF+n}{n}, \quad L_j = \binom{A_j}{U+1}.$$

Use ordinary expressions $S = EF$, $M = DS+1$, and natural B, K, r, V satisfying

$$SC = B + MK, \qquad M = B + r + 1, \qquad N(L_1, \ldots, L_m, R_D) = BV.$$

There are exactly $2m+9$ exposed coordinates. The complete conditional count is

$$2m + 9 + 2\nu_F + \nu_P + (m+1)\nu_B,$$

versus $2m+11+2\nu_F+2\nu_P+(m+1)\nu_B$ for the preceding interface under the same convention. The saving is exactly $\nu_P + 2$, but no numerical values for these uncompiled graph costs have been asserted. This is a proved compiler simplification, not yet a smaller closed ordinary consistency polynomial. Further quartic flattening and internal formalization have separate costs and proof obligations.

10 Comparison and what has not been achieved

The completed native family is a synthesis of previously separate tools, not a claim that each ingredient was invented here. The new mathematical obligations are their simultaneous canonical completeness, the ordering of sign and rank recovery, and the exact inverse reconstruction after each coordinate deletion. A superficially similar substitution can fail if it uses a canonical bound on arbitrary signed inputs, or if it forgets an integrality condition. The proof and independent checkers address these specific risks.

The lower-variable integer projections keep the same six fixed payloads. They do not claim reduced maximum expanded coefficient height or a minimum arithmetic circuit. Their definitions are factored where expansion is unhelpful; those registers still introduce no unknowns. Historical integer points include 25 variables at degree 920,26 at degree 214, and 24 at degree 1936. The new low-degree integer tradeoffs should be compared with those domains, not with the existing25-positive sextic or20-positive octic. Ten unrestricted variables and nine positive variables already exist at much larger degrees, so our variable counts are not absolute records.

The 11-positive degree277 construction improves both degree and gate count relative to the inspected325/113 predecessor. Its centered companion is a separate degree/cost tradeoff. Attempts to merge a fourth Pell root and reach ten low-degree positive unknowns still encounter preservation of the main root and transport residue channels. Radical rationality alone would not repair that obstruction. No such ten-variable theorem is claimed.

The direct compiler optimizations preserve the existing max-square pairing and zero-test semantics. A different Cantor coding could make pairing itself quadratic with no private auxiliary coordinate, but changing a whole proof checker requires a recursive code translation and a correctness theorem. That possibility is not a drop-in replacement for the current frozen checker. The CRT interface similarly still requires complete polynomial graphs for its remaining special functions. Neither local node counts nor uncompiled graphs are counted as final consistency-polynomial sizes.

11 Reproducibility and the formal proof boundary

The release contains editable TeX, compiled PDF, literal/factored equation specifications, complete coefficient bindings, source pins and inherited proof texts, normal and optimized finite checks, and independent audits. Original input ZIPs are preserved for authentication and replay. Run `python3 verify.py` in an extracted release; add `--replay-originals` to rerun the original incoming verifiers as well. Run `python3 negative_controls.py` for the root integrity controls. The replay works in an isolated temporary copy and compares selected rebuilt artifacts byte for byte. It does not decide whether any consistency equation has a solution.

There are three separate assurances. The exact algebra checks certify the finite displayed transformations and literal artifacts. The accompanying unbounded arguments establish soundness and completeness of the new steps. The inherited compiler/Pell/refutation interfaces provide the semantic calibration to finite ZFC proofs. This packet does not newly formalize their entire composition or export an encoded object-language ZFC biconditional. The current upstream formal developments are reported with their own scope; no local diagnostic is used to upgrade an upstream claim to a fresh kernel check.

Source references. The project is Vladimir Reshetnikov's ProveIt repository, https://github.com/VladimirReshetnikov/ProveIt/tree/main/Computability/DiophantineConsistency. Full immutable hashes, archive Git blob IDs and SHA256 values are recorded in the pin manifests. The reviewed comparison is https://github.com/VladimirReshetnikov/ProveIt/blob/2605669a532342f153b13eb6b4a0af4b54f5fa37/Computability/DiophantineConsistency/PROPOSAL_REVIEW_WAVE10.md. Authored proofs identify the relevant original archive and theorem at each import boundary. All copied sources retain their notices and provenance.